

Final Problem Statement (Route A: fields $\rightarrow S_{11}$, with UQ)

Goal. Learn a neural operator that maps microstrip antenna *geometry + materials + boundary/port conditions + frequency* to the complex time-harmonic E-field on a structured grid; derive $S_{11}(f)$ by port-mode overlap at a fixed Z_0 and reference plane; calibrate uncertainty to support robust *yield optimization* under fabrication tolerances.

Operator choice & physics hooks.

- Use a *Fourier Neural Operator (FNO)* backbone for resolution-invariant operator learning on the Yee grid (spectral layers plus a local term). FNO supports mesh/discretization transfer and zero-shot super-resolution on PDE solution maps.
- For complex time-harmonic fields and port quantities, either (A) keep the FNO real-valued with Re/Im channels, or (B) add a *complex-valued operator head* (e.g., DeepONet-style) for frequency-parametric transfer and boundary-aware post-training with a PEC tangential-**E** penalty.
- If/when physics regularization is desirable, borrow *operator-level* physics-informed terms (curl-curl residual on interior points and tangential-**E** penalties on PEC masks) to stabilize extrapolation without abandoning the operator-learning setting.

Design-space economy for yield. Constrain the surrogate domain using *response-variability Essential Directions (ED)*: compute in-band sensitivities of $|S_{11}|(f)$ at a nominal design, extract an orthonormal basis ordered by variability, then define a domain elongated along the top N_s directions (typically $N_s \in \{2, 3\}$) and tight elsewhere. This preserves yield-relevant axes while cutting data cost.

Inverse design & UQ. Train the forward surrogate (fields $\rightarrow S_{11}$). Perform *robust inverse design* in the ED domain to maximize spec yield $Y(x)$ under process tolerances. Use *band-event conformal calibration* on the surrogate’s S_{11} to convert predictive dispersion into calibrated pass probabilities, and verify with EM Monte Carlo on a held-out subset.

Assumptions to watch (with mitigations).

1. *Spectral truncation smoothing.* High- Q notches can be over-smoothed by low mode counts; mitigate via Fourier-mode sweeps, frequency-edge reweighting, and (if necessary) a small high-frequency branch near the port slice.
2. *Locality of ED.* ED is a first-order, local reduction. If optimization leaves the trusted region or topology changes, refresh gradients and rebuild ED (trust-region ED).
3. *Calibration under frequency shifts.* Calibrate on *band events* (e.g., $\max_{f \in B} |S_{11}| \leq \text{spec}$) rather than pointwise magnitude to maintain reliability when resonances shift.

Key References (titles only)

- *Fourier Neural Operator for Parametric Partial Differential Equations* (Li et al., 2021).
- *Learning the Solution Operator of Parametric PDEs with Physics-Informed DeepONets* (Lu et al.).
- *Essential Directions for Surrogate-Based Robust Antenna Design* (Kozieł et al., Sci. Rep., 2022).